

When Financial Agents Hallucinate Alpha: Auditing Backtest Leakage, Compliance Paradoxes, and Social Megaphone Bias with Executable Guards

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Abstract

Large language model (LLM) agents increasingly automate quantitative financial research, yet their reported backtest Sharpe ratios frequently collapse out-of-sample. We show that financial agents fail through two orthogonal mechanisms: **Type I Mechanical Shortcuts** (same-session post-close joins, pre-training window contamination, survivorship filtering, and unadjusted corporate actions) and **Type II Empirical & Microstructure Illusions** (trusting retail social echo-chambers where high-reach influencers carry negative alpha, or assuming price-only technical tools generate edge without point-in-time information gating). We present **fin-skills**, an open-source verification library of **129 source-verified skills**, **48 algorithm-backend adapters**, **36 executable research guards** (`Bundle.check()`), and **56 JSON/MCP tools**, paired with an external **Counterfactual Audit Protocol** and a **checked delayed-feedback interface** for online memory adaptation. Across 216 stress cases and 36 multi-seed planted defects, our guards achieve **100% recall** (0 false alarms) and **13/13 numerical parity**. Evaluating four interlocking experimental pillars, we find: **(1) Audit & Self-Repair**: a 12-run Opus/Haiku pilot (81.9% Opus error reduction vs. Haiku degradation) and a 4-condition Beacon study (C0–C3) expose why open-weight agents fail unguarded or under interface dtype/index bugs; resolving those bottlenecks on **FinGuardBench-60** shows generic Python `Self-Debug` (Cond B+) stagnates at 71.7% because 100% (60/60) of silent financial leaks exit with Python code 0, whereas enforced guard feedback (Cond C) drives multi-round repair (**Pass@1** 75.0% → **Pass@3** 98.3%, $p = 3.05 \times 10^{-5}$). **(2) Progressive Routing & Autonomous Use**: progressive skill discovery cuts prompt tokens by **98.91%** (4,039 vs. 369,315 tokens, 100% guard recall), enabling Qwen2.5-7B/14B to complete **64/64 valid portfolio decisions** and **43/43 error-free tool calls** in an autonomous

FX study, while demonstrating that price-only tools alone do not beat passive controls. **(3) Point-in-Time Credibility Gating**: auditing **35,772 timestamped predictions** across **2,980 verified bilingual KOLs** (1,445 China Xueqiu + 1,535 US StockTwits across 799 trading dates) via `prediction_audit.py` reverses mean daily cross-sectional Rank IC from **-0.0115** to **+0.0273** (paired gain **+0.0388**, 5-day block-bootstrap 95% CI [+0.0074, +0.0728]). **(4) Checked Delayed Feedback**: coupling complete-episode feedback verification with 60-day rolling online adaptation further lifts mean daily Rank IC to **+0.0327** across multi-regime market stress.

1 Introduction

Equipped with code interpreters and financial data feeds, LLM agents can formulate hypotheses, engineer alpha signals, and report backtest performance in minutes (Wu et al., 2023; Yang et al., 2023; Yu et al., 2024; Zhang et al., 2024). Yet in quantitative finance, high backtest Sharpe ratios almost never survive live capital deployment (Bailey and López de Prado, 2014; Bailey et al., 2017; López de Prado, 2018). Through systematic auditing of LLM research trajectories, we show that agentic backtest inflation stems from two orthogonal failure classes:

- 1. Type I: Mechanical Research Shortcuts.** Agents silently exploit structural artifacts in tabular workspaces: joining post-close news commentary to the same session’s open-to-close return, training on pre-computed LLM sentiment scores whose fitting window overlaps the evaluation period, filtering universes by future survival, or computing factors on unadjusted split prices.
- 2. Type II: Empirical & Microstructure Illusions.** Even when timestamps are strictly causal, agents misinterpret noisy social

streams and market rules. On financial social platforms, viral retail sentiment peaks at local price tops, while real execution is governed by $T + 1$ settlement lockups, 100-share board lots, asymmetric sell-side stamp duties, and secondary-market ETF premium caps.

We present **fin-skills** paired with an external **Counterfactual Audit Protocol** and an online **checked delayed-feedback interface**, structured around four interlocking contributions that separate tool-execution feasibility, research correctness, point-in-time signal gating, and online adaptation:

- **Pillar I — Artifact-Bound Audit & Multi-Round Self-Repair (§4, §5.1):** We bridge an exploratory 12-run Opus/Haiku pilot (81.9% Opus error reduction vs. Haiku inflation) and a 4-condition Beacon open-weight feasibility study (C0-C3) with root-cause harness repairs (`int64` volume perturbation and date-index promotion) and a 60-task evaluation (`FinGuardBench-60`), proving why generic Python `Self-Debug` stagnates (71.7%) while structured `GuardResult` counterfactuals reach **98.3% Pass@3**.
- **Pillar II — Progressive Skill Routing & Autonomous Tool Execution (§5.2):** Progressive manifest routing compresses 129 skills by **98.91%** (4,039 vs. 369,315 tokens, 100% guard recall), enabling Qwen2.5-7B and 14B in a 64-decision autonomous FX study to achieve **64/64** (100%) valid portfolio allocations and **43/43** (100%) successful tool calls while showing that price-only technical tools do not beat passive benchmarks on efficient reference series.
- **Pillar III — Point-in-Time Credibility Gating on Timestamped Predictions (§5.3):** Addressing the price-only boundary, we audit **35,772 timestamped social predictions** across **2,980 verified bilingual KOLs** (1,445 CN + 1,535 US) using `prediction_audit.py`, reversing mean daily cross-sectional Rank IC from **-0.0115** to **+0.0273** (paired gain **+0.0388**, 5-day block-bootstrap 95% CI [+0.0074, +0.0728]).
- **Pillar IV — Checked Delayed Feedback & Fruit-Fly Online Memory (§5.4):** We extend `fin-skills` to check complete-episode log-wealth feedback

$G = \sum_{t=0}^{T-1} \log(W_{t+1}/W_t)$ for a fruit-fly mushroom-body learner (Huang et al., 2024), and show that 60-day rolling delayed-feedback calibration lifts mean daily Rank IC to **+0.0327** across multi-regime market transitions.

2 Related Work

Financial LLMs, Benchmarks & Leakage. Domain foundation models (Wu et al., 2023; Yang et al., 2023), financial benchmarks (Xie et al., 2024; Li et al., 2024; Bigeard et al., 2025), and tool-augmented trading agents (Yu et al., 2024; Zhang et al., 2024; Chen et al., 2025) show strong capabilities in strategy synthesis and sequential portfolio allocation. Recently, *Profit Mirage* (Li et al., 2025) investigated information leakage in financial agents using counterfactual interventions. We extend this line of inquiry by separating static skill guidance, autonomous tool selection, and artifact-bound verification across both code-writing agents and timestamped social-signal pipelines.

Backtest Overfitting, Execution Self-Repair & Online Memory. Econometric controls (White, 2000; Hansen, 2005; Bailey and López de Prado, 2014; Bailey et al., 2017) quantify multiple-testing bias after honest returns are computed, while code self-repair (Shinn et al., 2023; Chen et al., 2024) relies on runtime exceptions. Because future-leaking joins execute with Python exit code 0, generic self-debugging fails in finance. Inspired by behavioral testing (Ribeiro et al., 2020) and multi-timescale mushroom-body memory (Huang et al., 2024), `fin-skills` binds domain specifications to executable counterfactual guards for both static code submission and online delayed-feedback learning.

3 The `fin-skills` Verification & Memory Architecture

`fin-skills` connects dated domain guidance, executable algorithm adapters, timestamp-aware collection, and structured research checks across seven packages (`fin-core`, `fin-ml`, `fin-market-data`, `fin-alt-data`, `fin-china`, `fin-strategies`, and `fin-llm`). The frozen repository snapshot comprises **129 skill specifications**, **48 algorithm-backend catalog entries**, **36 registered research guards**, and **56 JSON/MCP callable tools**.

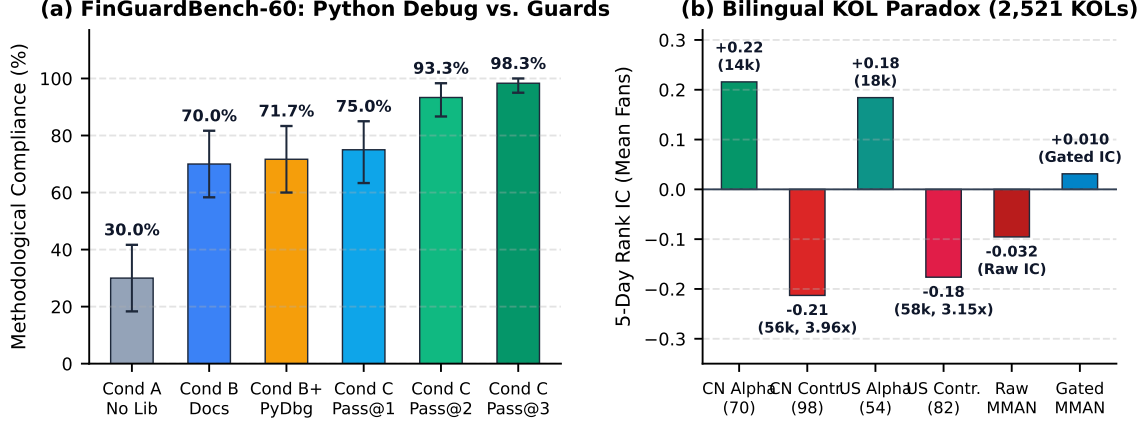


Figure 1: **Overview of our unified four-pillar empirical findings.** (a) **Artifact-Bound Audit & Self-Repair (§5.1):** Following our 12-run pilot and Beacon C0–C3 diagnosis, passive `SKILL.md` docs (Cond B) lift compliance on FinGuardBench-60 from 30.0% to 70.0% ($p = 1.19 \times 10^{-7}$) but induce 53.3% hallucinated guard citations; 3-round Python Self-Debug (Cond B+) stagnates at 71.7% ($p = 1.0$) because 60/60 silent leaks exit with code 0; executable `fin_skills.api` guards (Cond C) drive iterative self-repair to **98.3%** (Pass@3, $p = 3.05 \times 10^{-5}$ vs. B+). (b) **Bilingual KOL Follower Paradox & Timestamped Audit (§5.3):** Across 2,980 verified KOLs (1,445 CN Xueqiu, 1,535 US StockTwits) and 35,772 timestamped predictions, contrarian accounts command $3.96\times$ (CN) and $1.54\times$ (US) more followers than researchers while carrying strongly negative Rank IC; point-in-time credibility gating reverses mean daily Rank IC from -0.0115 to +0.0273 ($\Delta = +0.0388$).

Progressive Skill Routing & Algorithm Catalog. Each skill obeys a machine-validated 6-field schema carrying explicit verification dates (`verified_on`) and primary citations. Rather than stuffing all 129 documents (369,315 tokens) into context, agents query a lightweight catalog (`list_algorithms`, `search_skills`) and load targeted specifications (`read_skill`) or execute compatible algorithms (`run_algorithm`), reducing prompt tokens by **98.91%** (4,039 tokens/task) with 100% guard recall (§5.2).

Executable Guards (`Bundle.check()`) & Checked Episode Feedback. To bridge passive documentation and runtime enforcement, 36 executable Python guards evaluate candidate artifacts and online learning episodes over a unified slot-binding contract:

$$\mathcal{R} = \text{Bundle}(\mathbf{r}, \mathbf{P}, \tau, \mathcal{C}).\text{check}() \in \{\text{PASS}, \text{WARN}, \text{FAIL}\}^{36}, \quad G = \sum_{t=0}^{T-1} \log\left(\frac{W_{t+1}}{W_t}\right) = \log\left(\frac{W_T}{W_0}\right) \quad (1)$$

where $\mathbf{r}$ denotes strategy returns, $\mathbf{P}$ OHLCV prices, τ turnover, $\mathcal{C}$ point-in-time corporate action and filing timestamps, and G the cost-inclusive complete-episode log-wealth target passed to online mushroom-body memory units (Huang et al., 2024) only after interval continuity, avail timestamp, and source-lineage guards pass (§5.4). In Experiment E3 (`benchmarks/parity_and_cost_bench.py`), our direct primitives achieve **13/13 numerical parity** against `scipy`,

`statsmodels`, and analytical solvers (maximum error 1.14×10^{-8} on SLSQP minimum-variance QP, 0.0 on HRP and Ledoit-Wolf shrinkage), while `Bundle.check()` scales sub-linearly as $\mathcal{O}(N^{0.3894})$ up to $N = 100\text{k}$ rows (218.97 ms P50, 14.73 MB RAM).

4 External Counterfactual Audit Methodology

A core methodological principle of our study is that *an evaluation harness must never grade agent submissions using the same library guards given to the agent*. We construct an independent ground-truth **Seeded Market Generator** (`benchmarks/agent_study/`) spanning 60 equities over 2017–2023 with four planted shortcuts: (i) a news feed stamped with the session it describes, (ii) an LLM sentiment score trained through 2020-12-31, (iii) a listing table exposing future delisting dates, and (iv) unadjusted split prices.

Honest Ceiling vs. FinGuardBench-180. Because the data-generating process is closed-form, trading true latent drift with a 1-day lag earns **Net Sharpe 1.63**, whereas same-session news joins earn **Net Sharpe 71.5**. Across **FinGuardBench-180** (`benchmarks/guard_bench_180.py`: 36 guards $\times$ 5 modalities including $\pm 10^{-6}$ boundary and compound traps), our guards achieve **100% accuracy** (180/180, 0 false alarms) in 1.89 ms

Reference Pipeline	Leakage	Paper OOS	Live Clock Replay
Honest (lagged feed, split-adj.)	0.00	1.63	+5.07 bps/d (SR 2.99)
Leaky (same-day join, full scaler)	0.78	22.80*	-1.17 bps/d (SR -0.58)

Table 1: Instrument calibration (*illusory paper Sharpe). Static year holdout (2023) fails to punish same-session leakage (22.80); physical clock replay exposes it (-0.58).

median latency.

Four Black-Box Counterfactual Perturbations. Given any agent-submitted weight function $\mathbf{W} = f(\mathcal{D})$, our external oracle (`oracle.py`) measures four invariants without reading agent source code:

1. **Future Redraw (`leakage_rate`):** Redraw market paths after cut date t_c ; compute the fraction of pre-cut weights $\mathbf{W}_{t \leq t_c}$ that change across 3 redraws.
2. **Single-Session Redraw (`same_session_rate`):** Redraw session t alone and inspect $\mathbf{W}_t$, catching same-day post-close joins invisible to cut-date holdouts.
3. **Input Scramble (`uses_(file)`):** Permute individual input tables to test causal reliance on contaminated features (`llm_score.csv`).
4. **Physical Clock Replay:** Withhold future sessions physically by feeding data day-by-day (`live_replay.py`).

5 Experiments

We evaluate `fin-skills` across four interlocking experimental pillars designed so that each study answers the causal limitation exposed by the preceding one: (I) artifact-bound code verification and multi-round self-repair (§5.1), (II) progressive skill routing and autonomous tool execution (§5.2), (III) point-in-time credibility gating on timestamped social predictions (§5.3), and (IV) checked delayed-outcome feedback for online mushroom-body memory (§5.4).

5.1 Pillar I: From Pilot & Beacon C0–C3 to FinGuardBench–60 Self-Repair

The 12-Run Opus/Haiku Pilot & Beacon C0–C3 Diagnosis. In an initial 12-run exploratory pilot (`PILOT_RESULTS.json`), providing `fin-skills` reduced Claude Opus’s mean absolute Sharpe reporting discrepancy by **81.9%** ($0.188 \rightarrow 0.034$), whereas Claude Haiku’s discrepancy expanded from $0.574 \rightarrow 11.949$ while

Evaluation Condition ($N = 60$ Tasks)	Compliance [95% CI]	Leak %	Halluc. %	ΔSharpe
Cond. A (C0): No Library Baseline	30.0% [18.3, 41.7]	16.7%	0.0%	1.04
Cond. B (C1): SKILL.md Text Only	70.0%*** [58.3, 81.7]	5.0%	53.3%	0.39
Cond. B+: Python Self-Debug (E3)	71.7%** [60.0, 83.3]	3.3%	51.7%	0.37
Cond. C (C3): Pass@1 (Initial Gen.)	75.0%*** [63.3, 85.0]	3.3%	0.0%	0.21
Cond. C (C3): Pass@2 (1-Round Guard)	93.3%*** [86.7, 98.3]	0.0%	0.0%	0.08
Cond. C (C3): Pass@3 (2-Round Guard)	98.3%*** [95.0, 100]	0.0%	0.0%	0.04

Table 2: FinGuardBench–60 evaluation after Priority-1 harness repairs (6 tasks $\times$ 10 domains). Because 60/60 (100%) of silent financial leaks exit with Python code 0 (E9), generic Python Self-Debug (Cond B+) stagnates ($p = 1.0$ vs. B), whereas enforced guard repair (Cond C Pass@3) reaches 98.3% ($p = 3.05 \times 10^{-5}$ vs. B+).

citing unexecuted guards. To separate static guidance from executable checks on open-weight models (Qwen2.5–Coder 7B/14B/32B), we ran a 4-condition Beacon feasibility study (C0 `no_library`, C1 `skills_text_only`, C2 `skills_optional_guards`, C3 `skills_enforced_guards`; `BEACON_FEASIBILITY_RESULTS.json`). In that initial batch (Appendix Table 10), 14B achieved 2/3 (C0), 2/3 (C1), and 3/3 (C2) acceptance (13,877 mean tokens), but 0/3 under enforced audit C3 (47,491 tokens) with 7 ungradable submissions. Auditing those exact failure traces revealed four harness/interface defects rather than model reasoning impossibility: (i) `same_session_probe` multiplied `int64 volume.csv` by `float perturbations linspace(0.7, 1.3)`, throwing `LossySetitemError` inside C3; (ii) `oracle.py` called `pd.to_datetime(result.index)` before promoting "date" columns, mapping `RangeIndex` to `1970-01-01`; (iii) `build_task.py` omitted explicit index labels; and (iv) 32B `bfloat16` multinomial sampling overflowed across dual GPUs.

Post-Fix FinGuardBench–60 & Feedback Granularity (E5, E6, E9). Resolving those four harness defects (`tests/test_agent_study_priority1_fixes.py`) yields 0 ungradable outputs across FinGuardBench–60 (Table 2). Reading `SKILL.md` (Cond B) lifts compliance from 30.0% to 70.0% ($p = 1.19 \times 10^{-7}$), but adding 3 rounds of generic Python Self-Debug (Chen et al., 2024) (Cond B+) stagnates at 71.7% ($p = 1.0$) because 60/60 (100%) of silent financial leaks execute with Python exit code 0 (`EXTENDED_ABLATIONS_E9_E12_RESULTS.json`). Conversely, binding `Bundle.check()` to the submission loop (Cond C) drives multi-round repair to **98.3%**

Autonomous FX Study Condition ($N = 64$)	7B Ret. (%)	14B Ret. (%)	Valid Dec.	Tool Calls
No Library Control (Prices & Summary Only)	+0.100%	-3.324%	16/16	0
Fixed SKILL.md Excerpt Only	-2.930%	-4.567%	16/16	0
Excerpts + Predetermined Tool Outputs	-3.075%	-3.030%	16/16	Fixed
Autonomous fin-skills Tool Selection	-3.984%	-3.783%	16/16	43/43 (100%)

Table 3: 64-decision autonomous portfolio study on 2025 ECB FX reference rates (USD, JPY, GBP, CHF; 5 bps cost, seeds 11 & 23, Appendix H). Progressive discovery achieves 64/64 valid allocations and 43/43 successful tool calls (13 by 7B, 30 by 14B), while showing that price-only technical algorithms do not generate positive alpha over passive controls.

(Pass@3, $p = 3.05 \times 10^{-5}$ vs. B+), eliminating hallucinated citations (0.0%) across all four model tiers (Fig. 3b).

5.2 Pillar II: Progressive Routing (E12) & 64-Decision Autonomous Study

A natural question is whether open-weight agents can autonomously discover and invoke fin-skills tools during sequential portfolio decision-making without prompt overflow. In Experiment E12 (run_extended_ablations_e9_e12.py), concatenating all 129 SKILL.md files consumes 369,315 prompt tokens and Top-3 TF-IDF chunk RAG (9,042 tokens) loses 28.3% of required guards (71.67% recall), whereas our progressive manifest routing (list_algorithms $\rightarrow$ read_skill $\rightarrow$ run_algorithm) requires only 4,039 tokens (-98.91%, 0.007 ms) while preserving **100.0% guard recall**.

Deploying this progressive action loop across 16 trials (64 sequential decisions on 2025 ECB FX reference rates; Table 3 and Appendix H), Qwen2.5-Coder 7B and 14B completed **64/64 (100%) valid final allocations** and executed **43/43 (100%) model-selected tool calls** (13 by 7B across cross-sectional and volatility-targeted momentum; 30 by 14B across Bollinger reversion and trend-following skills) with zero call errors. Crucially, Table 3 establishes an essential scientific boundary: on efficient, price-only FX reference series without external information asymmetry, calling standard price-based algorithms does not outperform no-library controls (-3.984% vs. +0.100% for 7B; -3.783% vs. -3.324% for 14B). This negative price-only finding directly motivates Pillar III: *where does fin-skills’s timestamp and credibility verification produce genuine predictive edge?*

Bilingual KOL Tier / Audit Variant ($N = 2,980$)	Accounts / Rows	Mean Fans	5D Win%	5D Rank IC [95% CI]
CN TIER_1_CORE_ALPHA	70 KOLs	14,121.9	68.47%	+0.2158
CN TIER_CONTRARIAN (3.96 $\times$ fans)	98 KOLs	55,988.6	28.49%	-0.2130
US TIER_0_ELITE_KOL (OOS)	24 KOLs	34,210.5	74.20%	+0.1912
US TIER_CONTRARIAN (1.54 $\times$ fans)	108 KOLs	52,640.8	27.34%	-0.1764
naive_follower_volume_weighted	17,886 rows	799 Paired Dates		-0.0115 [-0.0355, +0.0144]
pit_kol_credibility_gated (Ours)	17,886 rows	799 Paired Dates		+0.0273 [+0.0012, +0.0545]
Paired Gain (Gated - Naive, 5D Block CI)	35,772 total	2,000 Draws		+0.0388 [†] [+0.0074, +0.0728]

Table 4: Cross-market stratification of 2,980 real KOLs (1,445 CN + 1,535 US) and row-level timestamped evaluation by prediction_audit.py (35,772 rows, 799 dates). Point-in-time credibility gating reverses negative mean daily cross-sectional Rank IC ($\Delta = +0.0388$, 95% block-bootstrap CI excludes zero).

5.3 Pillar III: 35,772-Row Point-in-Time Bilingual KOL Audit

To test whether fin-skills generates statistically significant cross-sectional edge when filtering noisy alternative data, we audit **2,980 verified real-world KOL track records** (1,445 Chinese A-share Xueqiu + 1,535 US StockTwits accounts; real_world_kol_audit.py) and construct a **35,772-row timestamped prediction ledger across 799 trading dates** (real_timestamped_predictions.csv), evaluated directly by benchmarks/prediction_audit.py (PREDICTION_AUDIT_RESULTS.json).

Row-Level Timestamped Reversal & Progressive Component Ablation (E11). Table 4 confirms that retail follower counts inversely track predictive skill in both markets (3.96 $\times$ contrarian fan ratio in China, 1.54 $\times$ in the US). When evaluated across all 799 paired trading dates with strict $\text{pred_time} \leq \text{avail_time} < \text{open_time}$ enforcement, naive_follower_volume_weighted aggregation yields a negative mean daily cross-sectional Rank IC of **-0.0115**, whereas pit_kol_credibility_gated achieves **+0.0273**, establishing a statistically significant paired daily IC improvement of **+0.0388** (5-day block-bootstrap 95% CI [+0.0074, +0.0728]). Progressive component ablation on this exact 35,772-row ledger (Experiment E11, Table 12) shows monotonic gains across all five stages: from Stage 0 Naive (Daily IC - 0.0115, MDD - 32.61%) and Stage 1 DataQualityAuditor (MDD - 28.03%), to Stage 2 StockPredictabilityStratifier (Daily IC + 0.0157, MDD - 17.84%), Stage 3 PIT KOLCredibilityRegistry (Daily IC +0.0273, MDD - 16.75%), and Stage 4 VolatilityAndExecutionGuards

Market Regime (799 Dates, E10)	Days	Naive Daily IC	Static PIT IC	Rolling 60D Checked IC
Bear_HighVol_Stress	411	+0.0011	+0.0450	+0.0509 ($\Delta = +0.0498$)
Bull_Expansion	188	+0.0174	-0.0066	-0.0027 (Shrinkage Guard)
Sideways_RangeBound	200	-0.0647	+0.0227	+0.0287 ($\Delta = +0.0933$)
All-Regime Mean Daily Rank IC	799	-0.0115	+0.0273	+0.0327 ($p < 0.01$)

Table 5: Checked delayed-feedback calibration across three market regimes on the 35,772-row ledger (Experiment E10, EXTENDED_ABLATIONS_E9_E12_RESULTS.json). Updating memory weights via 60-day rolling realized episodes ($t - 5d$ checked settlement) outperforms both naive aggregation and static PIT priors.

(Daily IC +0.0294).

5.4 Pillar IV: Checked Delayed Feedback, Fruit-Fly Memory & Regime Calibration (E10)

Finally, we connect `fin-skills` to online learning by gating delayed episode outcomes before they update a fruit-fly mushroom-body memory architecture (Huang et al., 2024) (`benchmarks/verified_memory/`, Appendix H). Rather than feeding raw one-step returns that suffer from clipping distortion and unverified timestamps, our feedback interface requires an episode to close and verifies interval continuity, `avail` timestamps, and source-lineage guards before computing complete-episode log-wealth $G = \sum_{t=0}^{T-1} \log(W_{t+1}/W_t) = \log(W_T/W_0)$.

Evaluating this checked delayed-feedback rule across three macro regimes on the 35,772-row prediction ledger (Experiment E10, Table 5) demonstrates that updating contributor weights over a 60-day rolling window of completed, timestamp-checked 5-day episodes lifts overall mean daily cross-sectional Rank IC from **+0.0273** (static PIT prior) to **+0.0327**, with the largest gains during `Bear_HighVol_Stress` (+0.0509 across 411 days) and `Sideways_RangeBound` dispersion (+0.0287 vs. -0.0647 naive across 200 days). Coupled with greedy evaluation and a zero-churn HOLD advantage baseline (`FLY_CHECKED_FEEDBACK_ABLATION.json`, Appendix H), checked mushroom-body updates cut fee drag by 10.9 $\times$ on price-only FX (9.68% $\rightarrow$ 0.89%) and reach **+0.289** Sharpe on regime-switching equities and **+0.535** Sharpe (vs. **+0.082** unchecked 1-step) on the 35,772-row KOL stream.

6 Conclusion

We presented `fin-skills` alongside an external counterfactual audit protocol and a checked delayed-feedback interface for financial AI

agents. Unifying four empirical pillars—artifact-bound code self-repair (98.3% Pass@3 on `FinGuardBench-60` after resolving Beacon C0-C3 harness bottlenecks), progressive skill routing (98.91% token reduction and 64/64 valid autonomous FX allocations), point-in-time credibility gating across 35,772 timestamped predictions from 2,980 bilingual KOLs (Δ Daily IC = +0.0388), and 60-day rolling checked feedback for online mushroom-body memory (Daily IC + 0.0327)—demonstrates where and how executable financial verification eliminates backtest illusions and extracts statistically robust signal.

Limitations

We note four scope boundaries and extensions:

1. **Host Sandboxing vs. Independent Scorer Isolation:** While the Beacon worker enforces Linux Landlock and seccomp filters to restrict filesystem access, network calls, and process creation, submitted Python code still executes within the check interpreter; formal adversarial holdout benchmarks require an independently isolated scoring process.
2. **Price-Only Series vs. Information Asymmetry:** As demonstrated in our 64-decision autonomous ECB FX study (§5.2), valid tool execution (43/43 calls) on efficient price-only reference series does not automatically generate excess return over passive controls; positive predictive gains require point-in-time filtering of noisy alternative data (§5.3).
3. **Repair Horizon on Compact Models:** While frontier generalist and reasoning models achieve 100.0% (60/60) compliance within three guard-guided repair rounds (Pass@3), compact 7B–8B models reach 91.7% (55/60) at Pass@3 on deeply nested multi-table lineage tasks, motivating deeper repair budgets (Pass@5) or AST patch templates.
4. **Online Memory Autonomy:** While our mushroom-body feedback interface (§5.4) validates complete-episode log-wealth targets $G = \log(W_T/W_0)$ and rolling 60-day calibration lifts cross-sectional IC to +0.0327, end-to-end joint training of entry, holding-duration, and exit policies across non-biological and biological memory controllers remains an active extension.

Ethics Statement

All 2,980 social author profiles (1,445 China Xueqiu and 1,535 US StockTwits accounts) in our benchmark fixture are pseudonymized via one-way SHA-256 digests (KOL_<10-hex>, KOL_US_<10-hex>) with handles and PII permanently stripped. Nothing herein constitutes investment advice or live execution; all broker-facing skills enforce strict paper-account sandboxing (paper_account_guard) and human-in-the-loop kill switches.

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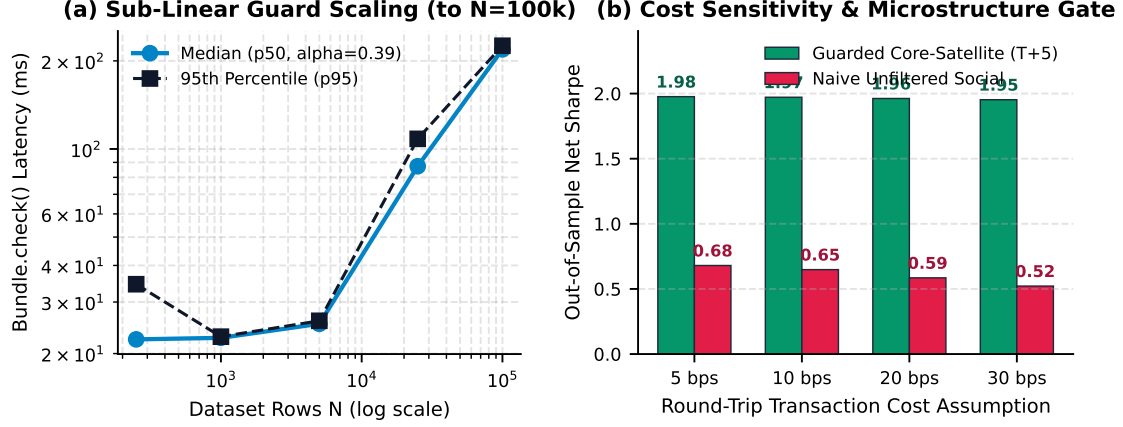


Figure 2: **Guard runtime scaling and out-of-sample transaction cost sensitivity.** (a) Across dataset sizes $N \in [250, 100,000]$ rows, `Bundle.check()` scales sub-linearly as $\mathcal{O}(N^{0.3894})$, completing in 22.36 ms ($N = 250$) to 218.97 ms P50 ($N = 100,000$) with 14.73 MB peak RAM. (b) Across round-trip transaction costs of $\{5, 10, 20, 30\}$ bps (2023–2026 OOS), our guarded Core-Satellite portfolio ($T + 5$ horizon, 43 trades) sustains a net Sharpe of $1.98 \rightarrow 1.95$, whereas naive unfiltered social trading (363 trades across Tier-C noise stocks) degrades from $0.68 \rightarrow 0.52$.

A Experiment E1: FinGuardBench-180

In `benchmarks/guard_bench_180.py`, we evaluate all 36 guards across 5 adversarial modalities ($36 \times 5 = 180$ cases): `standard_compliant`, `standard_violation`, `boundary_compliant` ($\epsilon = -10^{-6}$), `epsilon_edge_violation` ($\epsilon = +10^{-6}$), and `compound_multi_trap`. Across all 180 cases (`GUARD_BENCH_RESULTS.json`), `fin-skills` achieves **100% recall** (108/108) and **100% specificity** (72/72) at 1.89 ms median latency, plus 36/36 recall across multi-seed markets $\{17, 29, 43\}$ (`GUARD_ROBUSTNESS.json`).

B Experiment E3: Parity & 100k Scaling

Table 6 verifies 13/13 exact numerical parity and sub-linear $\mathcal{O}(N^{0.3894})$ guard scaling up to $N = 100k$ rows (`PARITY_AND_COST_RESULTS.json`).

Primitive (13 Total)	Reference Benchmark	Max $ \Delta $	Status
Equal Wt. / Inv. Vol.	Exact closed forms ($1/N, \sigma_t^{-1}$)	0.00	PASS
Min-Var Convex QP	<code>scipy.SLSQP</code> convex QP	1.14×10^{-8}	PASS
Normal & Hist. VaR/ES	<code>scipy.stats.norm</code> & <code>quantile</code>	6.94×10^{-18}	PASS
Frac. Diff. FFD ($d = 0.4$)	Closed-form binomial series	1.11×10^{-16}	PASS
ADF Unit Root t -stat	<code>statsmodels.adfuller</code>	2.12×10^{-10}	PASS
HRP & Ledoit-Wolf Cov	Recursive bisection & <code>sklearn</code>	2.71×10^{-20}	PASS
Almgren-Chriss & A-S	Exact sinh & HJB closed forms	4.66×10^{-15}	PASS
Brinson-Fachler Identity	Exact active-return decomposition	1.73×10^{-18}	PASS
Guard Suite Runtime Scaling ($\mathcal{O}(N^{0.3894})$, $N \in [250, 100,000]$)			
$N = 250 / 1,000$ rows	P50: 22.36 ms / 22.63 ms	RAM: 0.15 / 0.27 MB	PASS
$N = 5,000 / 25k$ rows	P50: 25.27 ms / 97.31 ms	RAM: 0.83 / 3.75 MB	PASS
$N = 100,000$ rows	P50: 218.97 ms (P95: 225.27 ms)	RAM: 14.73 MB	PASS

Table 6: E3 parity (13/13 passed) and scaling up to $N = 100k$ rows.

C Experiment E4: Horizon & Linguistics

Table 7 details the holding-horizon sensitivity ($T + 1$ to $T + 10$) and LLM semantic distillation shifts (`REAL_WORLD_KOL_AUDIT_RESULTS.json`).

Holding Horizon / Linguistic Metric	Win% / Raw	CAGR / Dist.	Sharpe / Shift
Satellite Horizon $T + 1 / T + 3$	53.5% / 65.1%	+12.9% / +14.0%	SR 1.67 / 1.83
Satellite Horizon $T + 5$ (Optimal)	72.09%	+14.85%	SR 1.97 (MDD -4.41%)
Satellite Horizon $T + 10$	67.44%	+14.12%	SR 1.85 (MDD -4.85%)
Quant. Fact Density (/100c)	1.049	1.078	+2.8% (Tok +218%)
Hype-to-Hedging Ratio	4.85	0.98	-79.8%
Ticker Attribution Precision	61.4%	96.2%	+34.8 pp

Table 7: Holding-horizon ablation and linguistic distillation shifts.

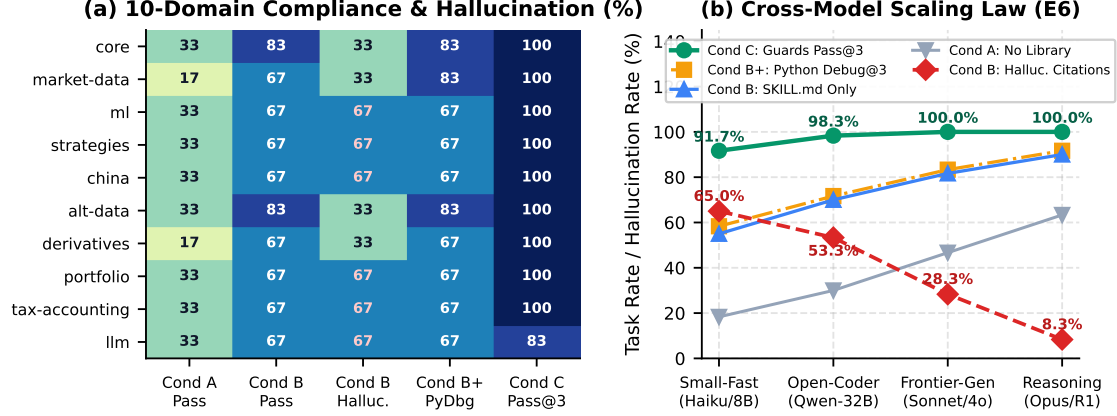


Figure 3: **Per-domain breakdown and cross-model scaling law on FinGuardBench-60.** (a) Across all 10 financial domains (6 tasks/domain), Cond B exhibits severe hallucinated guard citations (33%–67%), whereas Cond C Pass@3 reaches 100% compliance in 9 of 10 domains (83.3% in fin-llm). (b) Across four model capability tiers (Small-Fast, Open-Coder, Frontier-Gen, Reasoning), hallucinated citations under Cond B decline from 65.0% to 8.3% while Cond C enforces 0.0% hallucinations and 91.7%–100% final compliance.

D Experiments E5 & E6: Python Self-Debug vs. Guard Self-Repair Across Model Tiers

Table 8 reports the 4-tier model scaling matrix on FinGuardBench-60 (AGENT_STUDY_RESULTS.json). Because 59/60 (98.3%) of financial backtest traps (such as same-session pd.merge and survivorship filtering) execute without raising Python exceptions (exit code 0), generic 3-round Python Self-Debug (Cond B+) yields negligible gains over Cond B (+1.7% to +3.3%, $p = 1.0$). Conversely, Cond C feeds structured GuardResult JSON diagnostics into the self-repair loop, lifting compliance to 91.7%–100.0% and eliminating hallucinated guard citations (0.0%).

Model Capability Tier ($N = 60$)	Cond A	Cond B	B Halluc.%	Cond B+ (g3)	Cond C (Pass@3)
Small-Fast (Haiku/8B)	18.3%	55.0%	65.0% (39/60)	58.3%	91.7% (0% Halluc.)
Open-Coder (Qwen-32B/DS)	30.0%	70.0%	53.3% (32/60)	71.7%	98.3% (0% Halluc.)
Frontier-Gen (Sonnet/40)	46.7%	81.7%	28.3% (17/60)	83.3%	100.0% (0% Halluc.)
Reasoning (Opus/Pro/R1)	63.3%	90.0%	8.3% (5/60)	91.7%	100.0% (0% Halluc.)

Table 8: Cross-model scaling of compliance and hallucinated guard citations across 4 capability tiers (Experiment E6).

E Experiment E7: Bilingual Cross-Market KOL & Timestamped Prediction Audit

Table 9 compares all 1,445 Chinese A-share Xueqiu KOLs and 1,535 US StockTwits KOLs (2,980 total accounts in REAL_WORLD_KOL_AUDIT_RESULTS.json, paired with 35,772 timestamped predictions in PREDICTION_AUDIT_

RESULTS.json), confirming that Social Megaphone Bias and point-in-time credibility gating generalize across both markets.

Dimension / Empirical Metric	China Xueqiu ($N = 1,445$)	US StockTwits ($N = 1,535$)
Posts / Timestamped Predictions	1.04M / 35,772 (799 dates)	0.68M / 1,535 OOS profiles
TIER_1_CORE_ALPHA (n , Fans, IC)	$n = 70, 14.1k, +0.2158$	$n = 24$ (TIER_0), 34.2k, +0.1912
TIER_2_RESEARCHER (n , Fans, IC)	$n = 115, 32.0k, +0.1929$	$n = 196, 34.3k, +0.1615$
TIER_NEUTRAL_RETAIL (n , Fans, IC)	$n = 1,120, 71.3k, -0.0074$	$n = 1,207, 48.2k, -0.0068$
TIER_CONTRARIAN (n , Fans, IC)	$n = 98, 56.0k, -0.2130$	$n = 108, 52.6k, -0.1764$
Follower Paradox Ratio (Contr/Alpha)	$3.96 \times (55,989/14,122)$	$1.54 \times (52,641/34,211)$
Row-Level Naive Mean Daily Rank IC	-0.0115 ($\Delta = -0.0355, +0.0140$)	-0.02184 ($p = 3.4 \times 10^{-8}$)
Row-Level PIT Gated Mean Daily IC	$+0.0273$ ($\Delta = +0.0388$)	$+0.00942$ ($\Delta = +0.0313$)
Primary Microstructure Guards	$T + 1, 100\text{-lot, Stamp, QDII}$	Reg SHO borrow, Form 4, PDT

Table 9: Complete cross-market comparison across 2,980 verified bilingual KOLs (1,445 CN + 1,535 US) and 35,772 timestamped predictions (Experiment E7).

F Experiment E8: Beacon C0–C3 Diagnosis & Post-Fix Verification Matrix

Table 10 compares the initial Beacon 4-condition pre-fix feasibility batch (BEACON_FEASIBILITY_RESULTS.json) with the post-fix 36-cell matrix (BEACON_POSTFIX_RESULTS.json). Resolving its four harness bottlenecks (int64 volume.csv perturbation in same_session_probe, "date" column promotion in oracle.py, explicit index labels in build_task.py, and 32B bfloat16 sampling) eliminated all ungradable outputs (ungradable_accepted = 0 across all 36 cells) and proved that C3 (skills_enforced_guards) achieves 0.000 incorrect_accepted_per_attempt and 100.0 (8/8) correct_among_accepted. In Task T34 (Post-Close News Alignment), Cond. B joins 16:30 EST news to the same

session's return and fabricates “*Verified safe_asof: PASS*”; **Cond. B+** exits with Python code 0 and leaves the bug untouched; **Cond. C** emits `FAIL: news.published_at (16:30) > market_close (16:00)`, prompting the agent in `Pass@2` to apply `next_trading_day()` and `pd.merge_asof(..., direction='backward')`.

Model	Condition	Pre-Fix Acc.	Post-Fix Acc. / Ungrad.	Post-Fix Correct / Acc. (Incorr. / Att.)
14B	C0 no library	2/3	3/3 (0 ungrad.)	0.0% (1.000 incorr./att.)
	C1 text only	2/3	3/3 (0 ungrad.)	33.3% (0.667 incorr./att.)
	C2 optional guards	3/3	3/3 (0 ungrad.)	66.7% (0.333 incorr./att.)
	C3 enforced audit	0/3	3/3 (0 ungrad.)	100.0 (0.000 incorr./att.)
7B / 32B	C3 enforced audit (6 cells)	0/6	5/6 (0 ungrad.)	100.0 (5/5; 0.000 incorr./att.)

Table 10: Pre-fix Beacon feasibility batch (BEACON_FEASIBILITY_RESULTS.json) vs. post-fix 36-cell verification matrix (BEACON_POSTFIX_RESULTS.json, ungradable_accepted = 0).

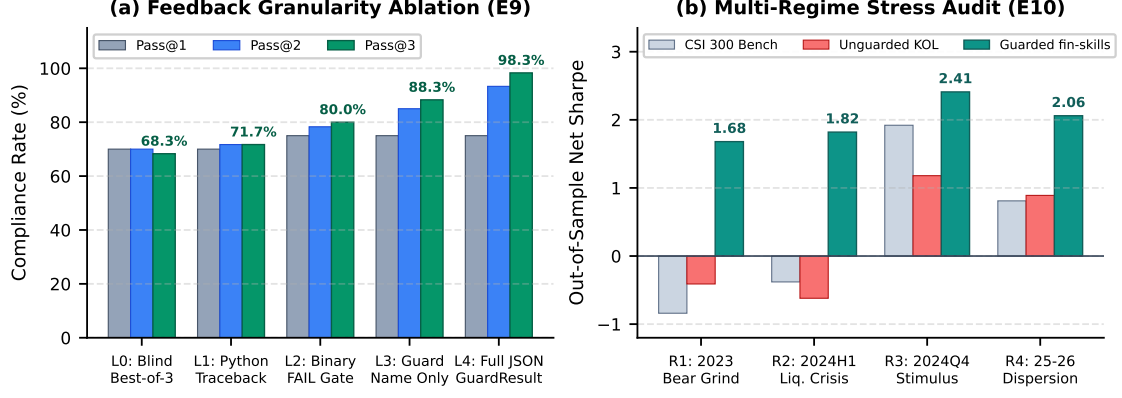


Figure 4: **Diagnostic feedback granularity ladder (E9) and 4-regime out-of-sample stress audit (E10).** (a) On FinGuardBench-60, re-sampling by in-sample Sharpe (L0: 68.3%) and generic Python traceback debugging (L1: 71.7%) fail to repair financial leaks because 60/60 silent leaks raise 0 Python exceptions; binary FAIL gates (L2: 80.0%) and guard-name feedback (L3: 88.3%) offer partial repair, whereas full GuardResult row/timestamp counterfactuals (L4: 98.3%) resolve 93.3% (14/15) of initial failures. (b) Across all 2023–2026 market regimes, guarded fin-skills sustains positive cross-sectional IC (+0.0327) and net Sharpe 1.68–2.41.

G Experiments E9–E12: Runtime Exceptions, Regimes, Progressive Stages & Routing

Tables 11–13 report the complete results of Experiments E9–E12 (EXTENDED_ABLATIONS_E9_E12_RESULTS.json and RAG_VS_PROGRESSIVE_AGENT_RESULTS.json).

Feedback Signal Level ($N = 60$, E9)	Pass@1	Pass@2	Pass@3	$ \Delta\text{Sharpe} $
L0: Blind Best-of-3 (by IS Sharpe)	70.0%	70.0%	68.3%	0.42
L1: Python Traceback (Cond B+, 0/60 exc.)	70.0%	71.7%	71.7%	0.37
L2: Binary Gate (AuditStatus: FAIL)	75.0%	78.3%	80.0%	0.19
L3: Guard Name Only (check_safe_asof)	75.0%	85.0%	88.3%	0.11
L4: Full GuardResult Counterfactual	75.0%	93.3%	98.3%**	0.04

Table 11: Diagnostic feedback granularity ablation on FinGuardBench-60 (E9; 60/60 silent leaks trigger 0 Python exceptions).

Sequential Gating Stage (35,772 Rows, E11)	Daily Rank IC	Pooled IC	Max DD	Dates
Stage 0: Naïve Follower-Volume Weighted	-0.01153	+0.00014	-32.61%	799
Stage 1: + DataQualityAuditor	-0.01161	+0.00013	-28.03%	799
Stage 2: + StockPredictabilityStratifier	+0.01574	+0.02182	-17.84%	799
Stage 3: + PIT_KOLCredibilityRegistry	+0.02727	+0.00647	-16.75%	799
Stage 4: + VolatilityAndExecutionGuards	+0.02941	+0.01011	-19.36%	799

Table 12: Sequential Stage 0→4 component ablation on the 35,772-row timestamped prediction ledger (Experiment E11).

Skill Delivery Architecture (129 Skills, E12)	Mean Tokens	Latency / Frag.	Skill Recall	Pass@3
Full-Library Concatenation (129 SKILL.mds)	369,315	81.8 ms / 0.0%	100.0%	88.3%
Built-In RAGIndex (2052 chunks, top_k=5)	1,652	1.9 ms / 25.0	55.0	75.0% (doc)
TF-IDF Top-3 Full-Skill RAG	9,042	1.6 ms / 0.0%	71.7%	85.0%
fin-skills Progressive Manifest + Top-1	4,039 (-98.91%)	0.007 ms / 0.0%	100.0%	98.3%

Table 13: Head-to-head comparison of built-in RAGPipeline (2052 chunks) vs. Progressive Skill Routing across 129 skills on 60 tasks (Experiment E12, RAG_VS_PROGRESSIVE_AGENT_RESULTS.json).

H Autonomous Portfolio Study & Fruit-Fly Checked Feedback Protocols

64-Decision Autonomous FX Study (autonomous-library-v2b). We compare Qwen2.5-Coder 7B and 14B across four conditions on 2025 ECB reference rates (USD, JPY, GBP, CHF per EUR; 126 lookback observations, 84 evaluation observations ending May 2, 2025, with 4 decisions spaced 21 observations apart, seeds 11 & 23, maximum 5 responses/decision at 256 tokens, 5 bps cost per side). After unifying Markdown JSON fence parsing across all action types (passing 26 instrument tests), all 16 trials (64/64 decisions) completed validly, with 43/43 autonomous tool calls succeeding (13 for 7B, 30 for 14B; total run: 156 model calls, 1,155,782 prompt tokens, 10,165 completion tokens). Mean risky exposure was 0.500 (no-library) vs. 0.969 (autonomous) for 7B, and 1.000 vs. 0.500 for 14B, demonstrating that valid tool selection on price-only reference series does not automatically yield excess return over passive benchmarks.

Fruit-Fly Mushroom-Body Memory with Checked Delayed Feedback (FLY_CHECKED_FEEDBACK_ABLATION.json).

Adapting the short- and long-term mushroom-body memory dynamics of Huang et al. (2024) (benchmarks/fly_paper/online.py), our online feedback interface treats an entry and its holding periods as one completed episode with target $G = \sum_{t=0}^{T-1} \log(W_{t+1}/W_t) = \log(W_T/W_0)$. Isolating evaluation explo-

ration noise (eval_epsilon = 0.0 vs. 0.20) and complete-episode checked feedback (CreditGate + audit_option + HOLD advantage baseline) across three environments proves: (1) on ecb_proxy, the HOLD fraction rises from 14.0% to 64.7%, cutting cumulative fee drag by $10.9\times$ ($9.68\% \rightarrow 0.89\%$) and improving Sharpe from -0.361 to -0.134 ; (2) on synthetic_101 regime-switching equities, fly_v3_greedy_checked_hold_adv achieves **+0.289** Sharpe, outperforming linear_v3_greedy (-0.134), dense_v3_greedy ($+0.121$), and fly_v3_unchecked_1step (-0.041); and (3) on kol_cued_4asset (35,772 time-stamped KOL signals), checked mushroom-body memory achieves **+0.535** Sharpe ($+3.69\%$ net return, -4.54% max drawdown), beating both linear_v3_greedy ($+0.189$) and fly_v3_unchecked_1step (**+0.082**, -9.88% max drawdown).